Consulting_Prj

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```
#load dataset
library(readr)
consult <- read_csv("Group3Data.csv")
```

```
## Rows: 39644 Columns: 61
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): url
## dbl (60): timedelta, n_tokens_title, n_tokens_content, n_unique_tokens, n_no...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
View(consult)

#check data channel via tech only
table(consult$data_channel_is_tech)
```

```
##
##     0     1
## 32298  7346
```

```
#Only 7346 obs came from tech out of 39,644 users/clients
#18.5% of the sample represented


#let new dataset contain only tech data by filtering
library(dplyr)
```

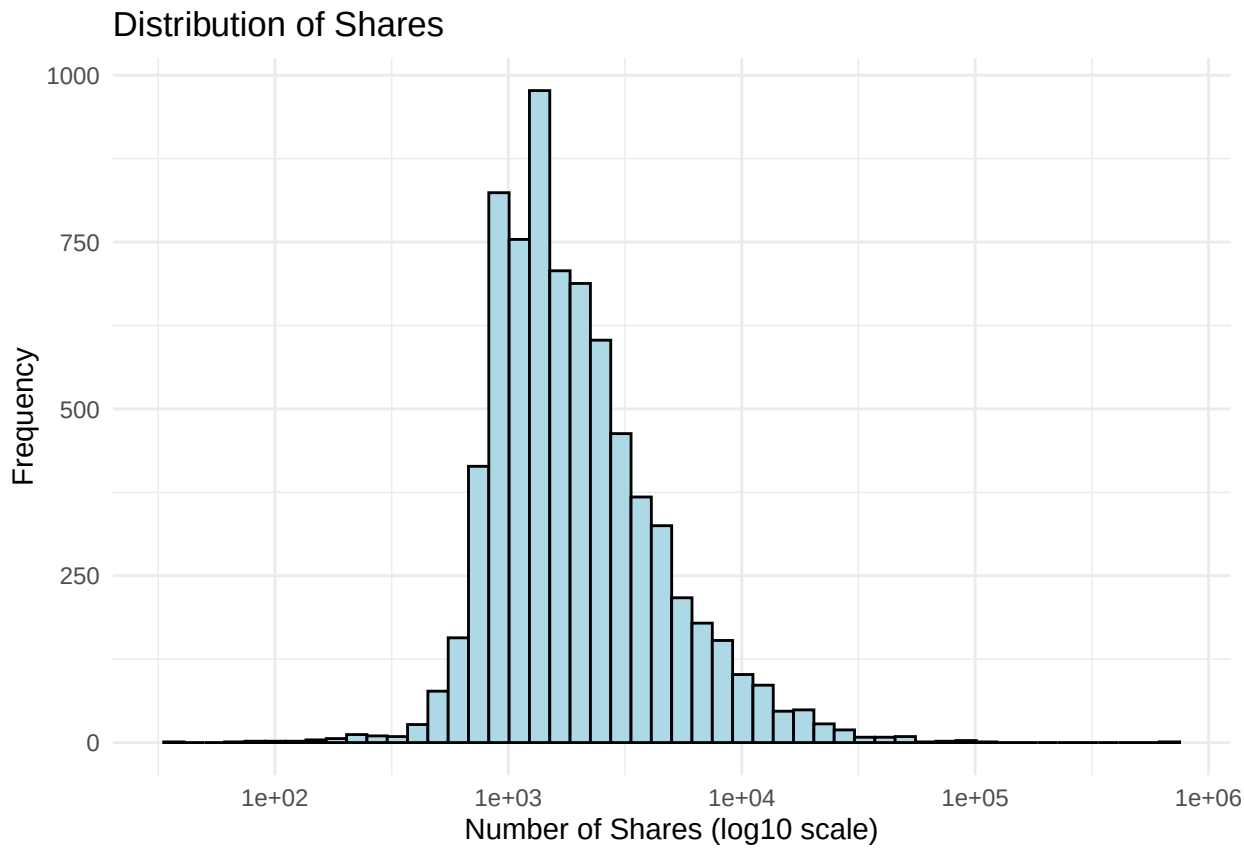
```
##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##     filter, lag
##
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
consult_tech <- consult %>%
  filter(data_channel_is_tech == 1)
```

```r
#-------------------------------------------------------------
#distribution of outcome variable (shares)
summary(consult_tech$shares)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      36    1100    1700    3072    3000  663600
```

```r
#visualization of shares
library(ggplot2)
ggplot(consult_tech, aes(x = shares)) +
  geom_histogram(bins = 50, color = "black", fill = "lightblue") +
  scale_x_log10() +
  labs(
    title = "Distribution of Shares",
    x = "Number of Shares (log10 scale)", y = "Frequency"
  ) +
  theme_minimal()
```




```r
#-------------------------------------------------------------
#combining weekdays into 1 variable
table(consult_tech$weekday_is_monday)
```

```
##
##    0    1
## 6111 1235
```

```r
library(dplyr)
consult_tech <- consult_tech %>%
  mutate(
    weekday = case_when(
      weekday_is_monday == 1 ~ "Mon",
      weekday_is_tuesday == 1 ~ "Tue",
      weekday_is_wednesday == 1 ~ "Wed",
      weekday_is_thursday == 1 ~ "Thur",
      weekday_is_friday == 1 ~ "Fri",
      weekday_is_saturday == 1 ~ "Sat",
      weekday_is_sunday == 1 ~ "Sun",
      TRUE ~ NA_character_
    )
  )

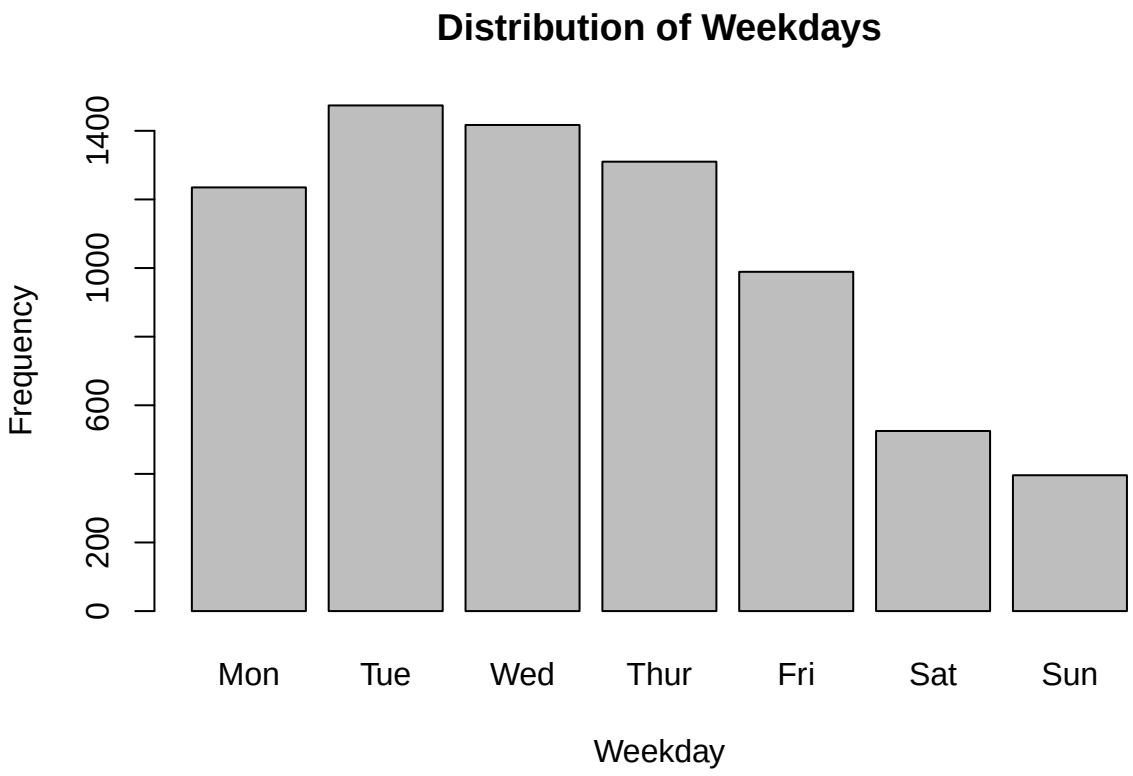

#make it a factor and order it
consult_tech$weekday <- factor(
  consult_tech$weekday,
  levels = c(
    "Mon", "Tue", "Wed", "Thur",
    "Fri", "Sat", "Sun"
  )
)



#see distribution
table(consult_tech$weekday)
```

```
##
##  Mon  Tue  Wed Thur  Fri  Sat  Sun
## 1235 1474 1417 1310  989  525  396
```

```r
#view plot
barplot(
  table(consult_tech$weekday),
  main = "Distribution of Weekdays",
  xlab = "Weekday",
  ylab = "Frequency"
)
```




```r
#grouping into weekday vs weekend
library(dplyr)
consult_tech <- consult_tech %>%
  mutate(
    day_type = case_when(
      weekday %in% c("Mon", "Tue", "Wed", "Thur", "Fri") ~ "Weekday",
      weekday %in% c("Sat", "Sun") ~ "Weekend",
      TRUE ~ NA_character_
    ),
    day_type = factor(
      day_type,
      levels = c("Weekend", "Weekday")
    )
  )

# Check distribution
table(consult_tech$day_type, useNA = "ifany")
```

```
##
## Weekend Weekday
##     921    6425
```

```r
#Polarity
summary(consult_tech$avg_negative_polarity)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
## -1.0000 -0.2850 -0.2220 -0.2288 -0.1667  0.0000
```

```
summary(consult_tech$avg_positive_polarity)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.3139  0.3560  0.3571  0.4013  1.0000
```

```
#Visuals
summary(consult_tech$num_imgs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   1.000   1.000   4.435   6.000  65.000
```

```
summary(consult_tech$num_videos)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000  0.0000  0.4472  1.0000 73.0000
```

```
summary(consult_tech$num_hrefs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   5.000   7.000   9.417  11.000 120.000
```

```
#sentiment
summary(consult_tech$global_sentiment_polarity)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -0.33912  0.09713  0.14379  0.14577  0.19102  0.65000
```

```
#-----------------------------------------------------------------
#descriptive plots
#shares by day
library(dplyr)

shares_by_day <- consult_tech %>%
  group_by(weekday) %>%
  summarise(
    n_articles = n(),
    total_shares = sum(shares, na.rm = TRUE),
    mean_shares = mean(shares, na.rm = TRUE),
    median_shares = median(shares, na.rm = TRUE)
  )

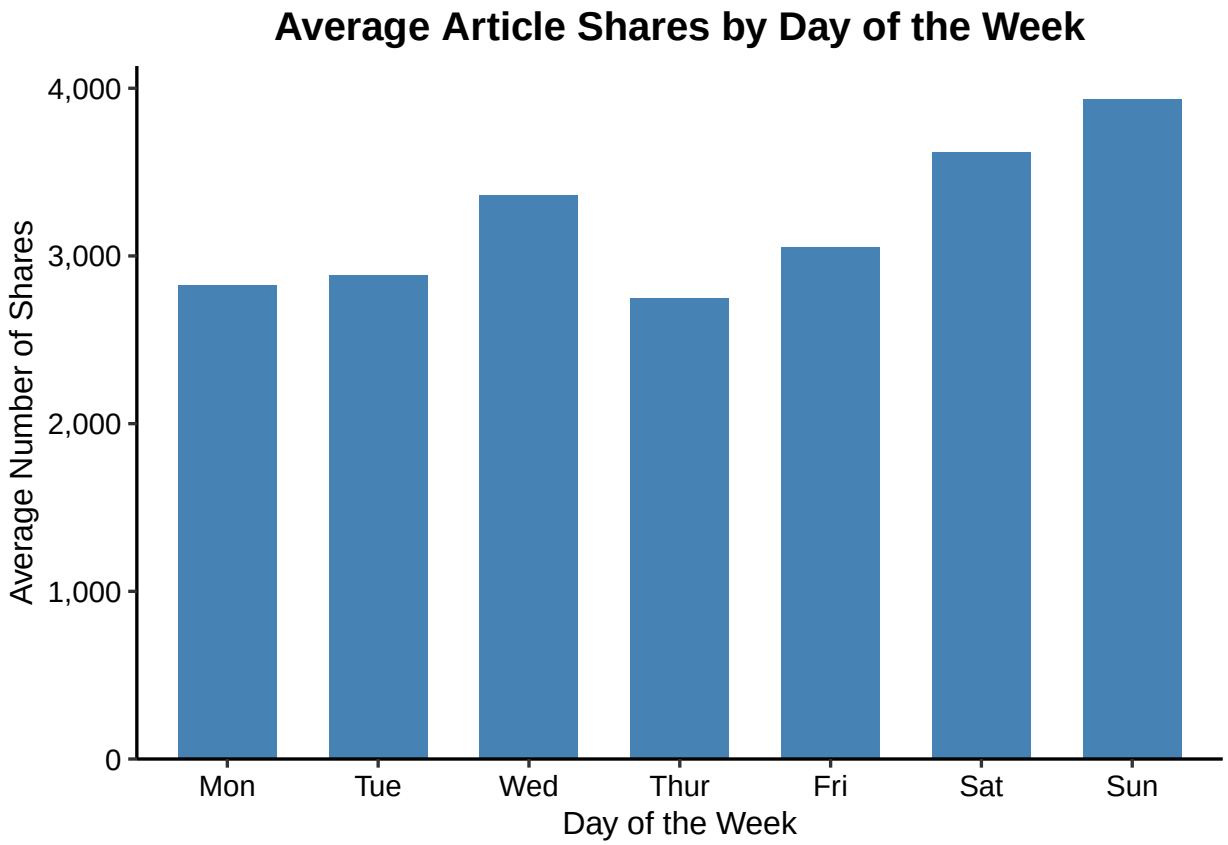
print(shares_by_day)
```

```
## # A tibble: 7 x 5
##   weekday n_articles total_shares mean_shares median_shares
##   <fct>        <int>        <dbl>       <dbl>         <dbl>
## 1 Mon           1235      3484532       2821.          1600
```

```
## 2 Tue           1474      4250146       2883.          1600
## 3 Wed           1417      4765065       3363.          1600
## 4 Thur          1310      3595351       2745.          1600
## 5 Fri            989      3017254       3051.          1800
## 6 Sat            525      1898113       3615.          2300
## 7 Sun            396      1558532       3936.          2400
```

```r
#visualization of shares by day
library(ggplot2)

ggplot(shares_by_day,
       aes(x = weekday, y = mean_shares)) +
  geom_col(fill = "steelblue", width = 0.65) +
  scale_y_continuous(
    labels = scales::comma,
    expand = expansion(mult = c(0, 0.05))
  ) +
  labs(
    title = "Average Article Shares by Day of the Week",
    x = "Day of the Week",
    y = "Average Number of Shares"
  ) +
  theme_classic(base_size = 13) +
  theme(
    plot.title = element_text(
      face = "bold",
      hjust = 0.5,
      size = 15
    ),
    axis.text.x = element_text(
      size = 11,
      color = "black"
    ),
    axis.text.y = element_text(
      size = 11,
      color = "black"
    ),
    axis.title = element_text(size = 12),
    panel.grid = element_blank()
  )
```




```
#heatmap correlation matrix between Number of links, Number of images,
#Number of videos and shares

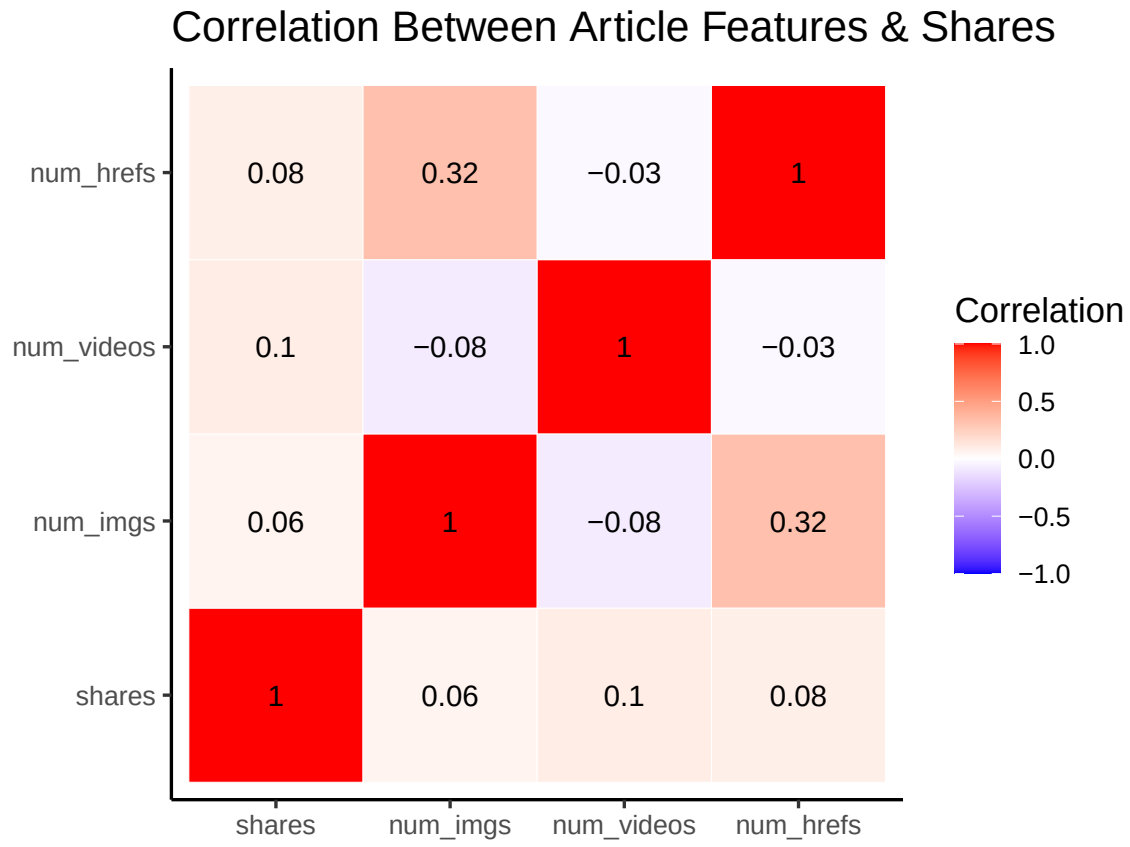
library(ggplot2)
library(dplyr)
library(tidyr)

cor_data <- consult_tech %>%
  select(shares, num_imgs, num_videos, num_hrefs) %>%
  cor(use = "pairwise.complete.obs",
      method = "spearman")

cor_long <- as.data.frame(as.table(cor_data))

ggplot(cor_long,
       aes(x = Var1, y = Var2, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Freq, 2)), size = 4) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    limits = c(-1, 1)
  ) +
  coord_equal() +
```

```
  labs(
    title = "Correlation Between Article Features & Shares", #spearman corr matrix
    x = NULL,
    y = NULL,
    fill = "Correlation"
  ) +
  theme_classic(base_size = 13)
```




```
#Regression with dummy variables
# Convert weekday to factor
consult_tech$weekday <- as.factor(consult_tech$weekday)
# Set Monday as reference
consult_tech$weekday <- relevel(consult_tech$weekday,ref = "Mon")
# Run regression
model_reg <- lm(shares ~ weekday, data = consult_tech)
summary(model_reg)
```

```
##
## Call:
## lm(formula = shares ~ weekday, data = consult_tech)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3730  -1907  -1345    -45 660237
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2821.48     256.72  10.991   <2e-16 ***
## weekdayTue     61.93     348.02   0.178   0.8588
## weekdayWed    541.30     351.20   1.541   0.1233
## weekdayThur   -76.94     357.82  -0.215   0.8298
## weekdayFri    229.33     384.97   0.596   0.5514
## weekdaySat    793.97     470.04   1.689   0.0912 .
## weekdaySun   1114.20     520.99   2.139   0.0325 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9022 on 7339 degrees of freedom
## Multiple R-squared:  0.001406,   Adjusted R-squared:  0.0005897
## F-statistic: 1.722 on 6 and 7339 DF,  p-value: 0.1114
```

```
#-------------------------------------------------------
# Set Sunday as reference
consult_tech$weekday <- relevel(consult_tech$weekday,ref = "Sun")
# Run regression
model_reg2 <- lm(shares ~ weekday, data = consult_tech)
summary(model_reg2)
```

```
##
## Call:
## lm(formula = shares ~ weekday, data = consult_tech)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3730  -1907  -1345    -45 660237
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3935.7      453.4   8.681   <2e-16 ***
## weekdayMon   -1114.2      521.0  -2.139   0.0325 *
## weekdayTue   -1052.3      510.6  -2.061   0.0394 *
## weekdayWed    -572.9      512.8  -1.117   0.2639
## weekdayThur  -1191.1      517.4  -2.302   0.0213 *
## weekdayFri    -884.9      536.5  -1.649   0.0991 .
## weekdaySat    -320.2      600.5  -0.533   0.5938
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9022 on 7339 degrees of freedom
## Multiple R-squared:  0.001406,   Adjusted R-squared:  0.0005897
## F-statistic: 1.722 on 6 and 7339 DF,  p-value: 0.1114
```

```
#-------------------------------------------------
# Run sensitivity analysis with weekday type
model_sensitivity <- lm(shares ~ day_type, data = consult_tech)

summary(model_sensitivity)
```

```
##
```

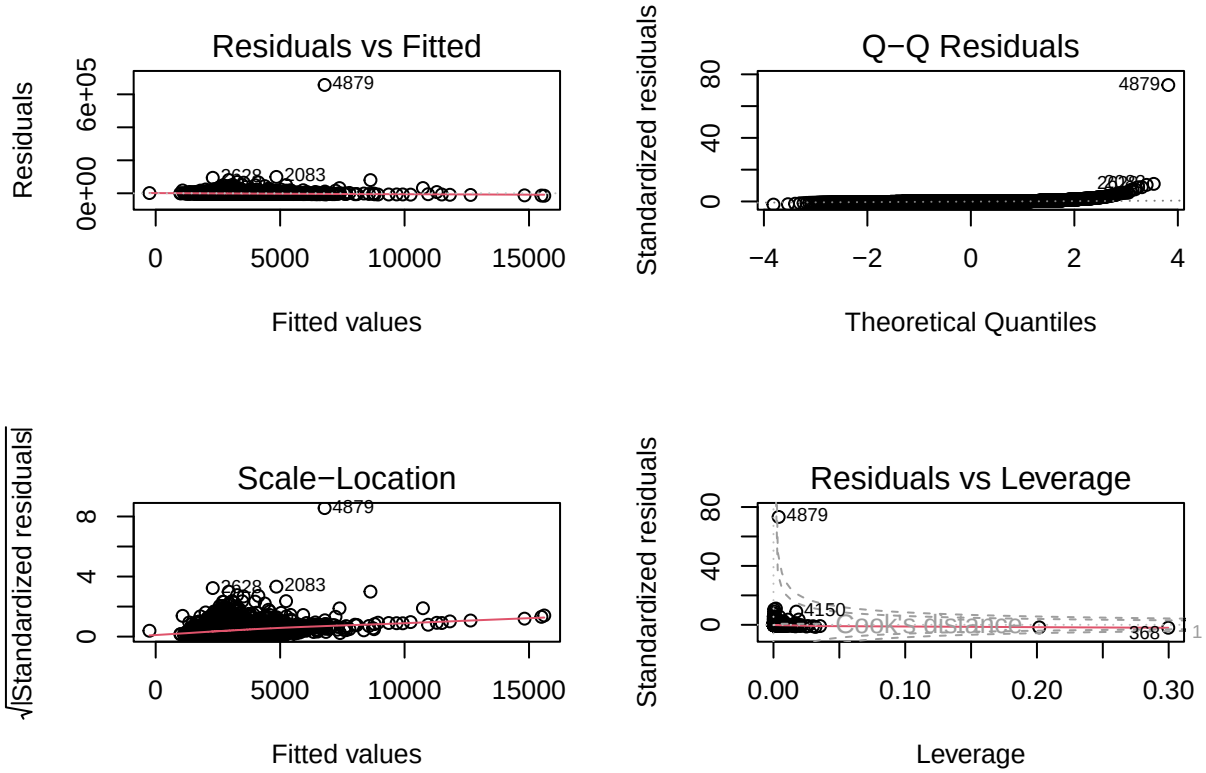
```
## Call:
## lm(formula = shares ~ day_type, data = consult_tech)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3634  -1875  -1375    -75 660625
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3753.1      297.3  12.626   <2e-16 ***
## day_typeWeekday   -778.5      317.9  -2.449   0.0143 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9021 on 7344 degrees of freedom
## Multiple R-squared:  0.0008161,  Adjusted R-squared:  0.00068
## F-statistic: 5.998 on 1 and 7344 DF,  p-value: 0.01434
```

```
#add more variables
model2 <- lm(shares ~ n_tokens_content + num_hrefs + num_imgs +
               num_videos + num_keywords +
               title_sentiment_polarity + global_subjectivity,
             data = consult_tech)

summary(model2)
```

```
##
## Call:
## lm(formula = shares ~ n_tokens_content + num_hrefs + num_imgs +
##     num_videos + num_keywords + title_sentiment_polarity + global_subjectivity,
##     data = consult_tech)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -14855  -1819  -1184     50 656813
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)              2375.6907   780.8749   3.042 0.002356 **
## n_tokens_content            1.0728     0.2781   3.858 0.000115 ***
## num_hrefs                  64.9042    14.8780   4.362  1.3e-05 ***
## num_imgs                  -56.6345    17.5815  -3.221 0.001282 **
## num_videos                140.8543    68.0879   2.069 0.038608 *
## num_keywords               -9.8131    62.8554  -0.156 0.875942
## title_sentiment_polarity  641.5667   450.5051   1.424 0.154459
## global_subjectivity      -697.8149  1401.7202  -0.498 0.618621
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8985 on 7338 degrees of freedom
## Multiple R-squared:  0.009711,   Adjusted R-squared:  0.008766
## F-statistic: 10.28 on 7 and 7338 DF,  p-value: 6.909e-13
```

```
#residual plot
par(mfrow = c(2, 2))
plot(model2)
```




```
par(mfrow = c(1, 1))
```

```
#investigate extreme observation
# Identify the observation with the largest residual
which.max(residuals(model2))
```

```
## 4879
## 4879
```

```
# Examine its characteristics
consult_tech[as.integer(
  names(which.max(residuals(model2)))
), ]
```

```
## # A tibble: 1 x 63
```

```
##   url                  timedelta n_tokens_title n_tokens_content n_unique_tokens
##   <chr>                    <dbl>          <dbl>            <dbl>           <dbl>
## 1 http://mashable.com~       274             13             2591           0.238
## # i 58 more variables: n_non_stop_words <dbl>, n_non_stop_unique_tokens <dbl>,
## #   num_hrefs <dbl>, num_self_hrefs <dbl>, num_imgs <dbl>, num_videos <dbl>,
## #   average_token_length <dbl>, num_keywords <dbl>,
## #   data_channel_is_lifestyle <dbl>, data_channel_is_entertainment <dbl>,
## #   data_channel_is_bus <dbl>, data_channel_is_socmed <dbl>,
## #   data_channel_is_tech <dbl>, data_channel_is_world <dbl>, kw_min_min <dbl>,
## #   kw_max_min <dbl>, kw_avg_min <dbl>, kw_min_max <dbl>, kw_max_max <dbl>, ...
```